Name : Kalaiyarasan S

Reg No : 222305549

Department : BSc Computer Science

SECURE CHAT

END-TO-END ENCRYPTION

Introduction

- **Project Overview:** A state-of-the-art web application engineered for complete data privacy.
- **Core Philosophy:** Cryptographic control belongs to the user, not the server.
- **Key Feature:** End-to-End Encryption (E2EE) using AES-256-GCM.
- **Real-time Power:** Powered by Socket.IO for instantaneous, bi-directional communication.

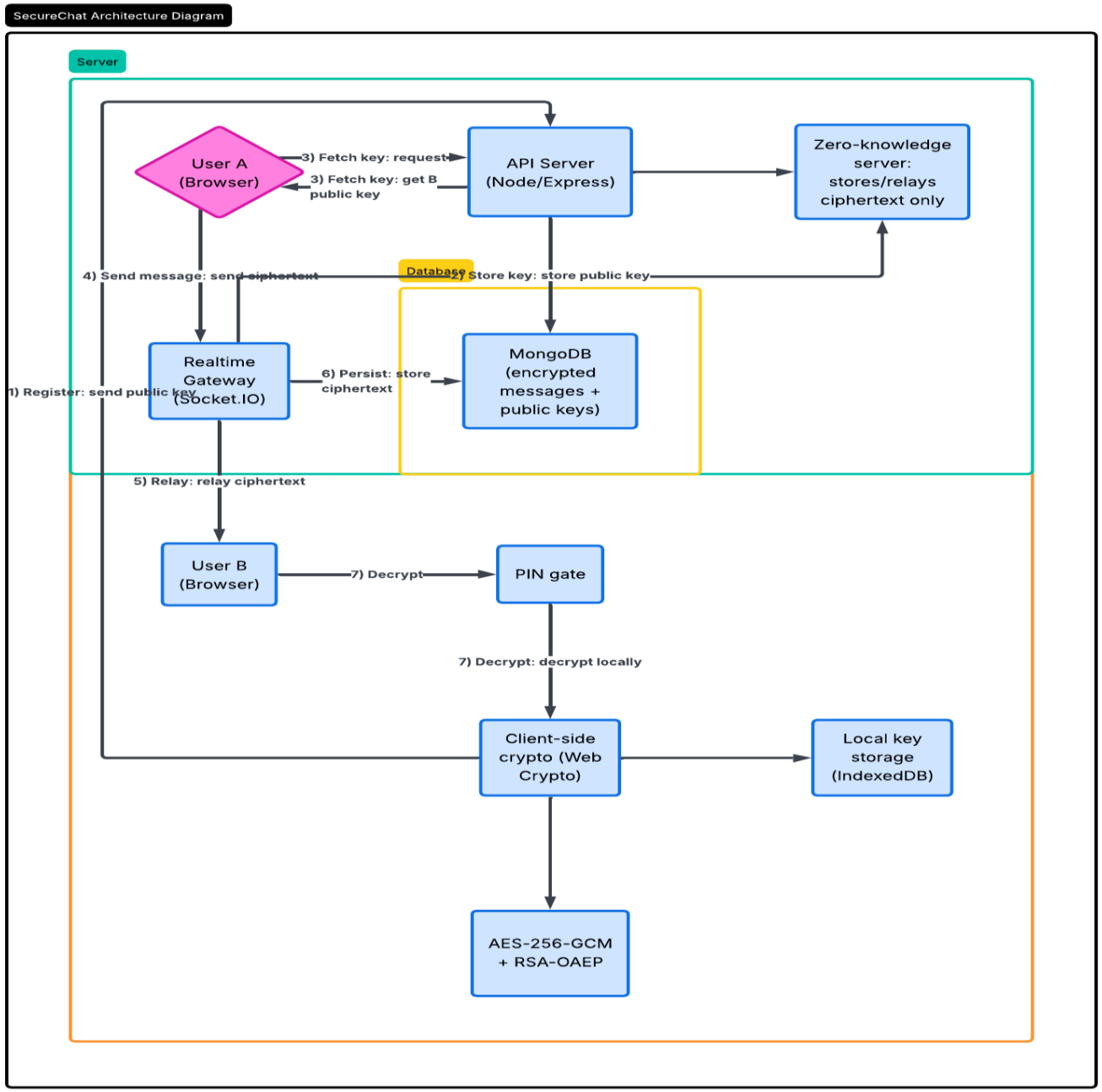
Problem Statement & Objectives

The Problem: Traditional platforms use server-side encryption where providers hold the keys, creating a **"single point of failure."**

Primary Objectives:

- Implement True E2EE: Encryption on the sender's device; decryption only on the receiver's.
- Zero-Knowledge System: The server never sees plaintext messages or private keys.
- User Empowerment: Establishing a foundational guarantee of privacy.

UML Diagram


SecureChat Architecture Diagram
Server
User A (Browser)
3) Fetch key: request
3) Fetch key: get B public key
API Server (Node/Express)
Zero-knowledge server: stores/relays ciphertext only
4) Send message: send ciphertext
Database
2) Store key: store public key
Realtime Gateway (Socket.IO)
6) Persist: store ciphertext
MongoDB (encrypted messages + public keys)
1) Register: send public key
5) Relay: relay ciphertext
User B (Browser)
7) Decrypt
PIN gate
7) Decrypt: decrypt locally
Client-side crypto (Web Crypto)
Local key storage (IndexedDB)
AES-256-GCM + RSA-OAEP

Module-wise Work Done

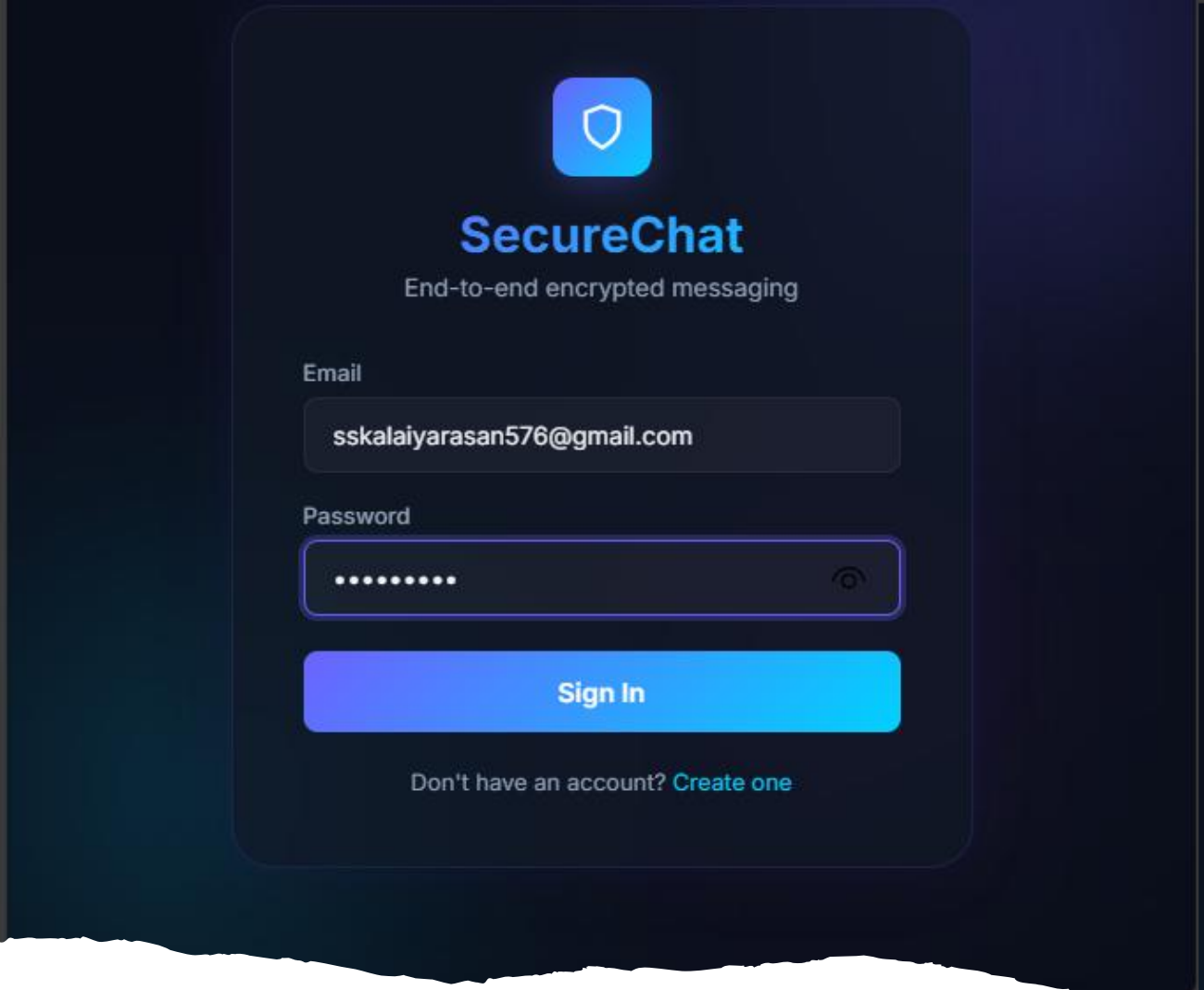
Frontend (React): Designed the responsive Chat UI.

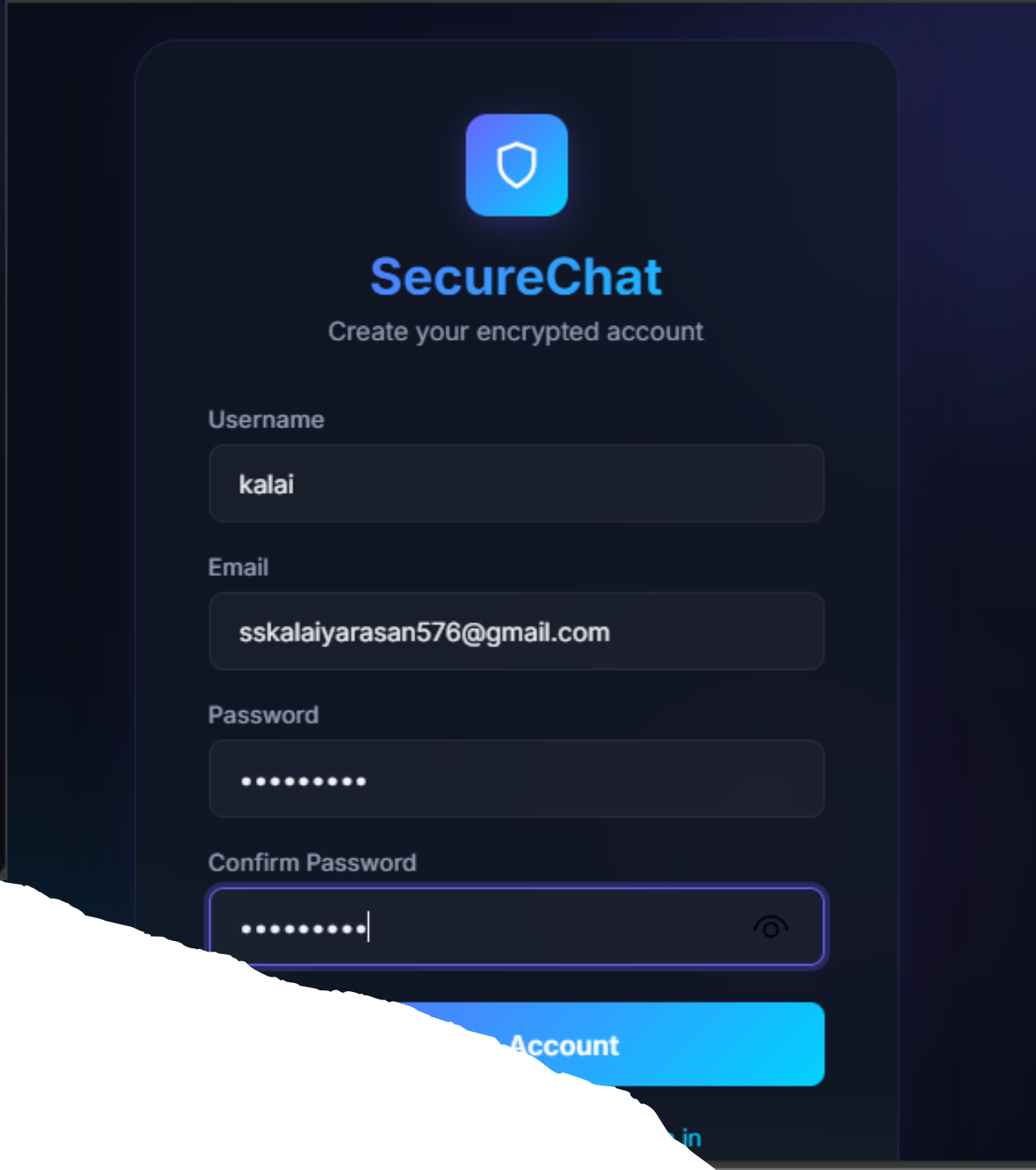
Integrated Web Crypto API for hardware-accelerated encryption.

Backend (Node.js/Express): Built the REST API for user registration and JWT-based authentication.

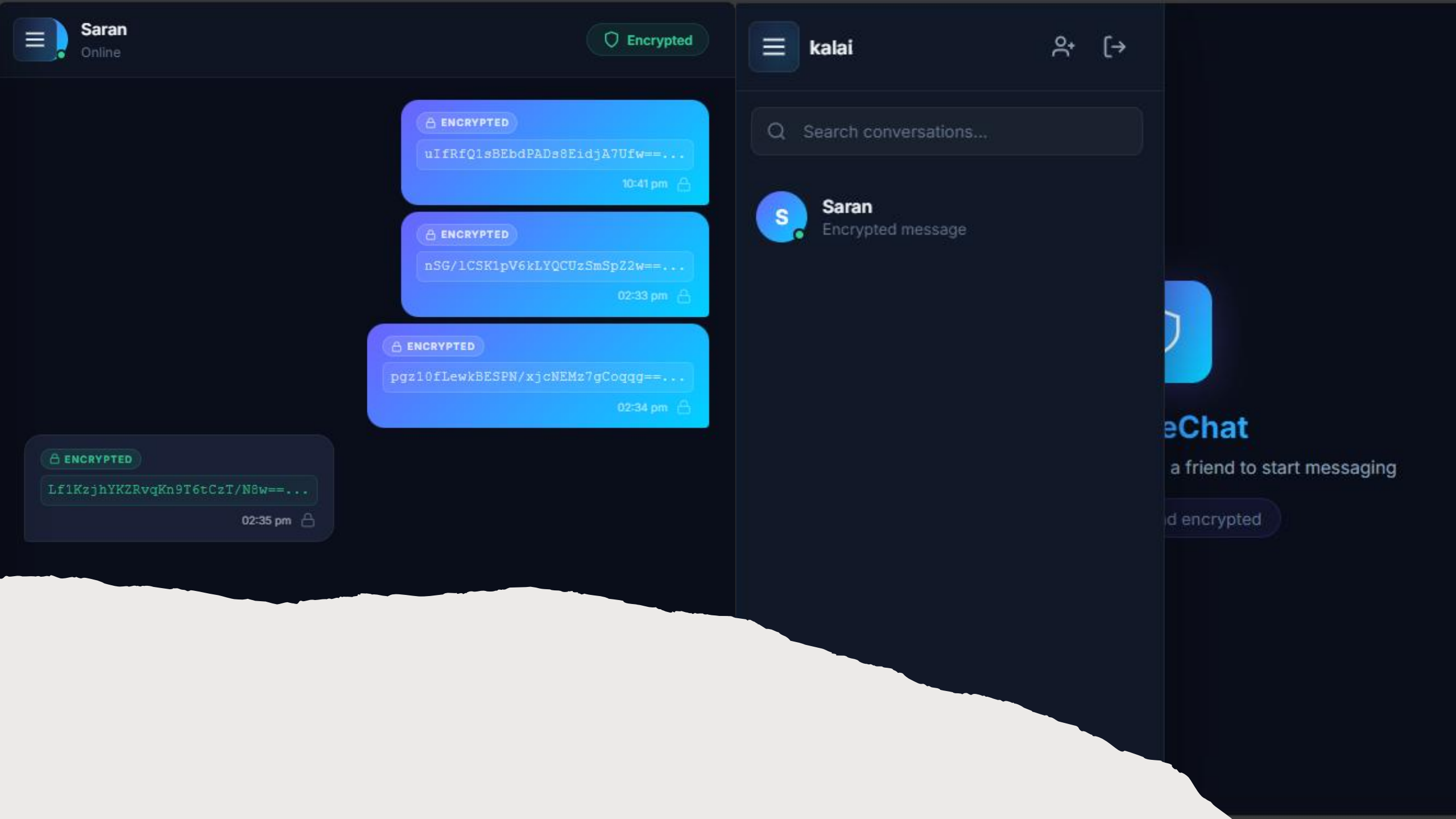
Managed active socket connections for real-time delivery.

Database (MongoDB): Schema design for Users and Message logs (storing ciphertext).





Screenshots


Saran
Online
Encrypted
ENCRYPTED
uIfRfQ1sBEbdPADs8EidjA7Ufw==...
10:41 pm
ENCRYPTED
nSG/1CSK1pV6kLYQCUzSmSpZ2w==...
02:33 pm
ENCRYPTED
pgz10fLewkBESPN/xjcNEMz7gCoqqg==...
02:34 pm
ENCRYPTED
Lf1KzjhYKZRvqKn9T6tCzT/N8w==...
02:35 pm
kalai
Search conversations...
S
Saran
Encrypted message
eChat
a friend to start messaging
d encrypted

kalai
Online

Decrypted

10:41 PM

DECRYPTED
hii
02:33 PM

DECRYPTED
hiiiii
02:34 PM

DECRYPTED
hai
02:35 PM

DECRYPTED
hii
05:50 PM

Type a message...

Enter Decrypt PIN

Enter your 4-digit PIN to view decrypted messages

Unlock

Review Record

Initial Review: Requirement gathering and selection of the MERN stack.

Mid-Term Review: Successful implementation of Socket.IO for real-time messaging.

Final Review: Integration of AES-256-GCM encryption and final security testing.

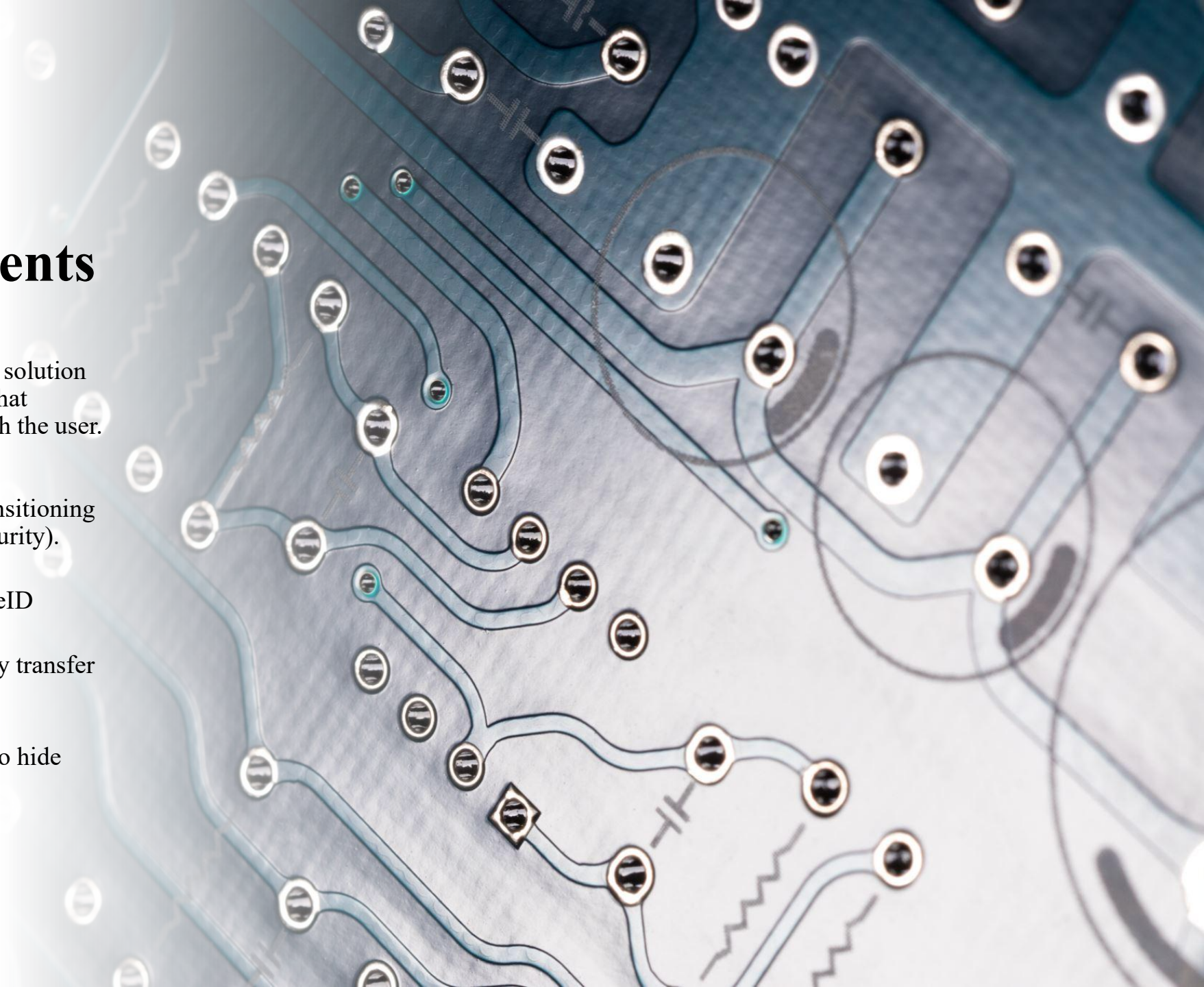
References

- **HTML/CSS**: MDN Web Docs
- **JavaScript**: JavaScript.info
- **Database**: MongoDB
- **Backend**: Express.js and Node.js

Conclusion
&
Future Enhancements

Summary: SecureChat provides a robust solution for private communication by ensuring that cryptographic control remains solely with the user.

- **Future Enhancements:**
  - **Group Chat Encryption**: Transitioning to MLS (Messaging Layer Security).
  - **Biometric Decryption**: Using WebAuthn for Fingerprint/FaceID access.
  - **Multi-Device Sync**: Secure key transfer via QR codes.
  - **Metadata Anonymization**: Implementing Onion Routing to hide communication patterns.



Thank You